MODULE 9

Dark Net and Deep Web Investigation

Certified Cyber Criminologist - Police / LEA Training

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What You Will Learn

The dark web is not magic - it's a network with rules. Learn the rules, and learn your limits.

- **Surface, deep and dark web** - clearly separated, no confusion ever again
- **How Tor works** - onion routing, three relays, three layers
- **Onion links and dark markets** - analyzed like an investigator
- **Tor artifacts** - what a seized device remembers

What You Will Learn (2)

Plus the law, the safety rules, and the discipline of real LEA work.

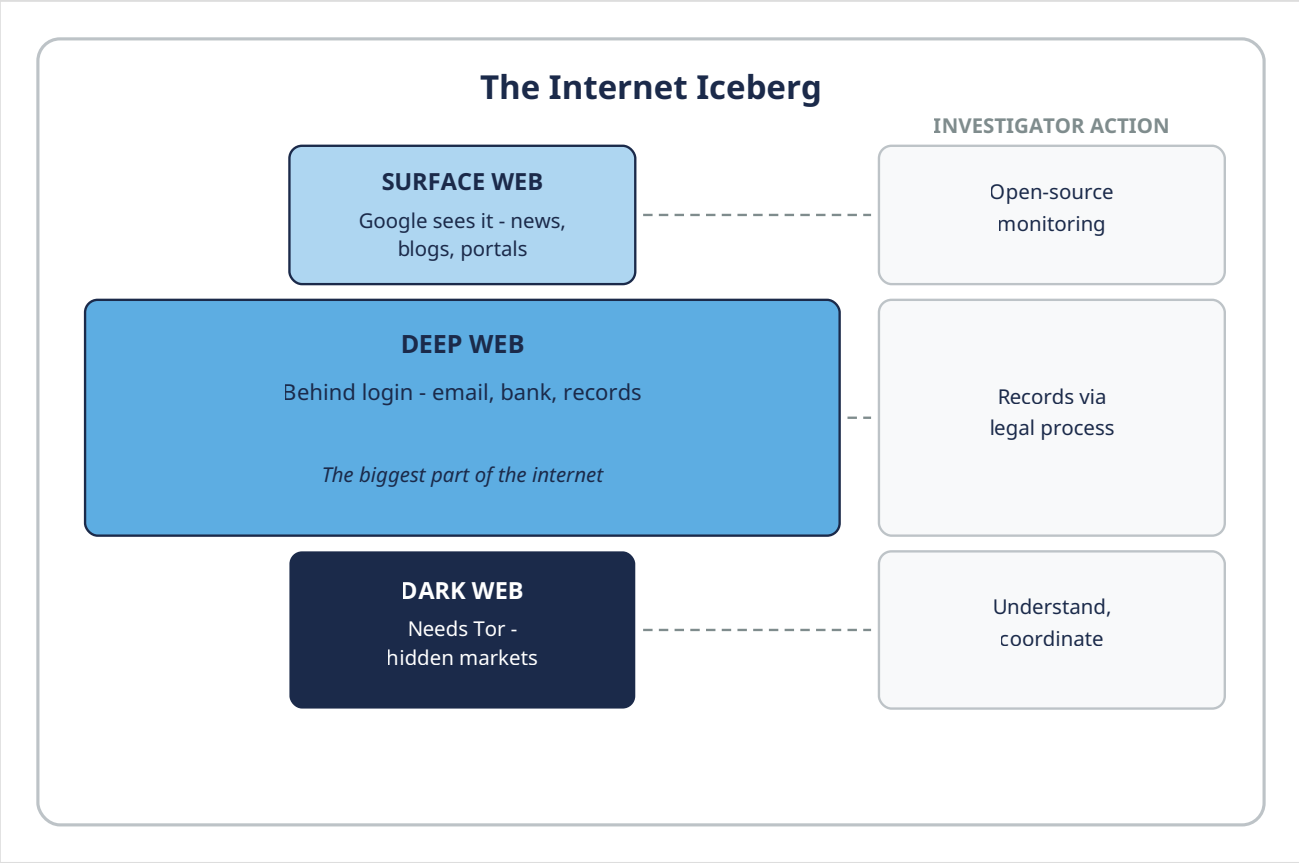
- **Indian laws** for dark-web cases: NDPS, IT Act, BNSS s.94, BSA s.63
- **OPSEC** - the six rules that keep YOU safe
- **Escalation red flags** - knowing exactly where your authority ends
- **Real cases** - Silk Road, NCB seizures, and what they teach

Is It Legal to Go There?

"Sir, what is the dark web?" "The internet that doesn't show on Google." "Sir, is it legal to go there?"

- **"Going there is legal.** Committing crime there is the biggest offence."
- "Going unprepared is **putting yourself in danger**."
- This module is the senior's full answer - **first understand, then lab**. Field work is for seniors only.

The Internet Iceberg



- **Surface web:** Google-visible - smallest part, open-source gold
- **Deep web:** behind logins - email, bank records; biggest, mostly legitimate
- **Dark web:** needs Tor - anonymity is the point; small, high crime risk

Three Webs: One Table, Zero Confusion

Memorize this table. It ends every confusion about "the dark web":

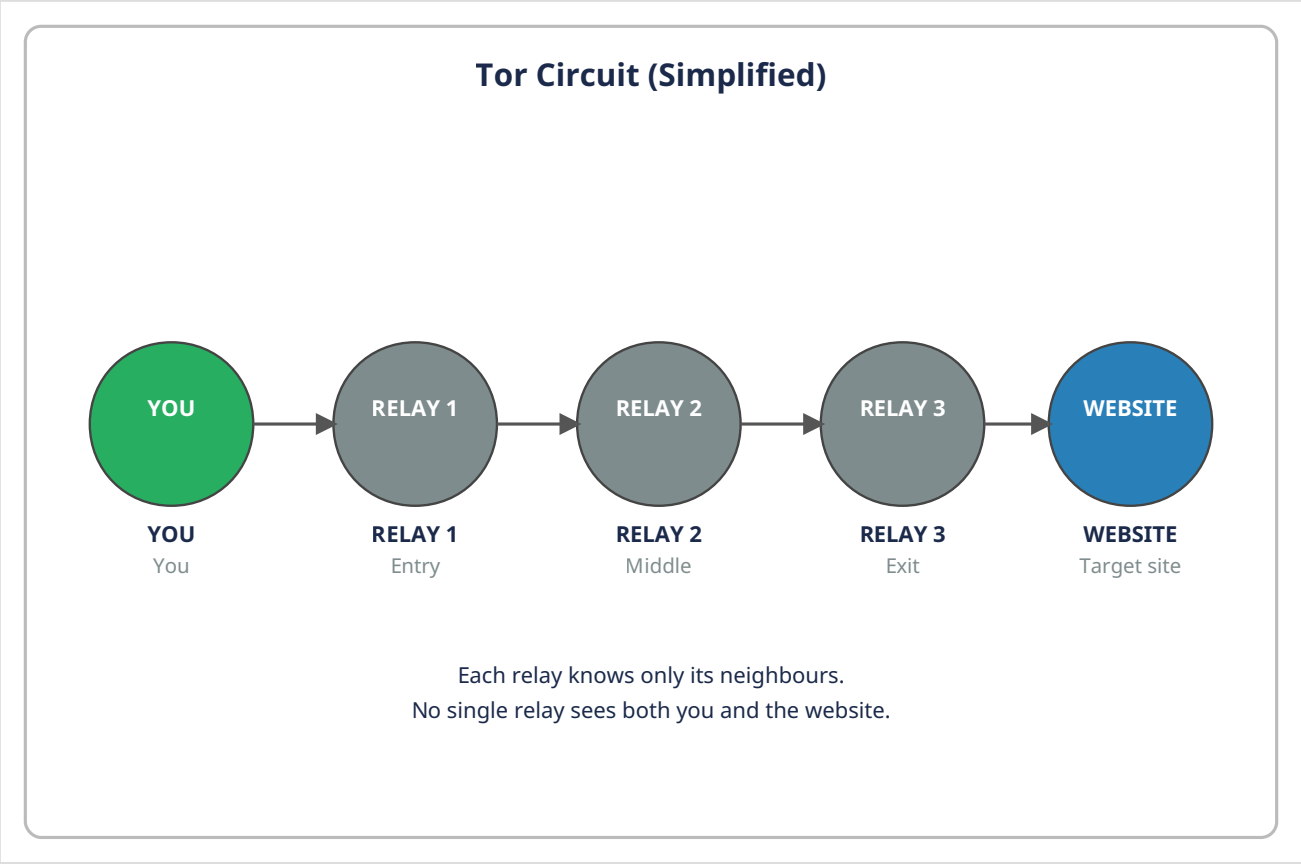
Web	Access	Example	Investigator action
Surface	Google search	News sites	Open-source monitoring
Deep	Login needed	Email, bank	Records via BNSS s.94
Dark	Tor software	Onion sites	Understand + coordinate

Deep Web: Where the Records Live

The deep web is where the RECORDS live - and records are what convict:

- Bank statements, CDR, email records, server logs - **all deep web**
- Get them through legal process: **BNSS s.94** (earlier CrPC s.91)
- Taking them without legal process is illegal - **and poisons your case**
- Deep web does NOT mean crime - it means login

Tor and Onion Routing



- Your traffic wraps in **3 encryption layers** - like envelopes inside envelopes
- Guard knows YOU, exit knows the DESTINATION, middle knows NEITHER
- No single relay sees the full picture - that's onion routing

The Three Relays: What Each Knows

Each relay opens only its own layer. Memorize what each one knows - and doesn't:

- **GUARD relay:** knows your IP address, does NOT know your destination
- **MIDDLE relay:** knows neither your IP nor your destination - just passes along
- **EXIT relay:** knows the destination website, does NOT know who you are
- An attacker needs **ALL THREE** - run by volunteers in different countries

Hidden Services and Onion Links

Dark websites hide their servers too - reaching one uses SIX relays, not three:

- Your 3 relays + the hidden service's 3 relays = **server location stays hidden**
- Addresses look strange: **expy3w4v...onion** - they open ONLY in Tor Browser
- Directory authorities publish relay lists; your browser picks 3 at random
- A new circuit per site makes tracking even harder

Analyzing an Onion Link

An onion link in a seized phone is the START of investigation, not the end. Check four things:

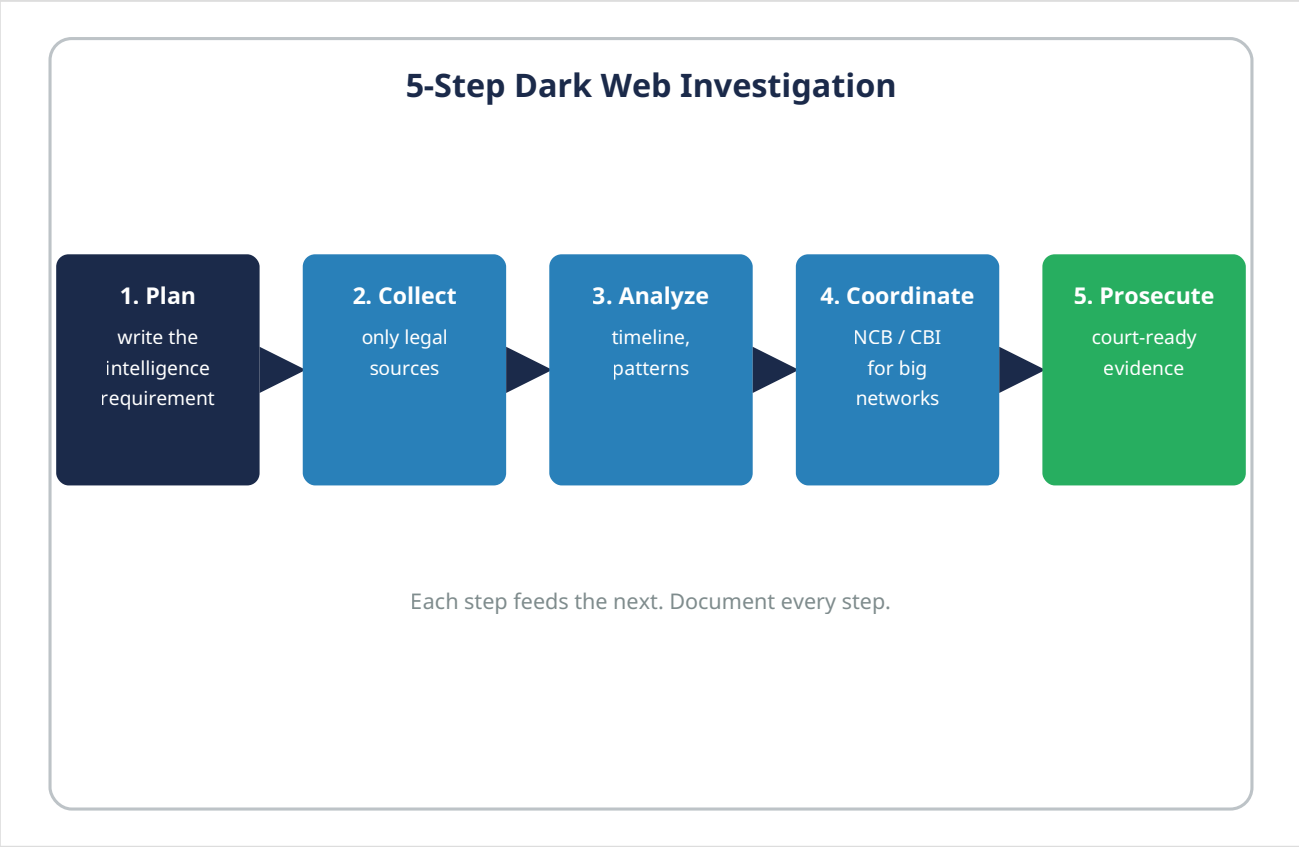
- **Age:** new site = untrusted; old durable site = a big gang behind it
- **Spread:** shared on 3 forums = wide reach and a real network
- **Activity:** 50 vendors, regular reviews = the market is alive
- **Language patterns:** same spelling mistakes as a dead site = same gang, new shop

Structure of Dark Web Markets

Dark markets run like e-commerce - understanding the system reveals the players:

- **VENDOR:** sellers with ratings and sales history - a profile is a dossier
- **ESCROW:** the market holds money till delivery - cash never goes direct to vendor
- **REVIEWS:** expose genuine vs fake vendors; one writing style across accounts = one person
- **FORUMS:** gang chatter and fraud complaints - old posts hold their mistakes

The 5-Step Investigation Workflow



- **Plan → Collect → Analyze → Coordinate → Prosecute**
- Every step's output is the next step's input
- Documentation runs alongside - not at the end

A Case's Journey: Parcels at the Post Office

Two drug parcels caught. Foreign return address. Watch the 5 steps work:

- **PLAN:** write the intelligence requirement - what to know, what NOT to do, whose permission
- **COLLECT:** parcel records (with approval) + NCB press releases on PIB - source-note everything
- **ANALYZE:** timeline + pattern chart - both parcels from the same foreign city
- **COORDINATE + PROSECUTE:** bigger than one city? Send NCB your full file, chain of custody intact

OPSEC: Your Safety First



- **Separate VM, separate identity** - your real self never touches the dark web
- Never download, never go alone, document everything
- Personal and official stay separate - always

OPSEC Rules in Detail

OPSEC - operational security - comes BEFORE any dark-web work:

- **Written permission** from your senior FIRST - alone, you endanger yourself and the case
- Official work on **official devices only** - your personal phone must never appear
- Real identity: no real email, name, or photo - **ever**
- Dark-web files can carry malware - **observe only, never download**

Tor Artifacts on a Seized Device

After a Tor session, the disk remembers. These artifacts tell WHEN Tor ran and HOW:

- **places.sqlite** (profile folder): modified time = when the session last ran
- **torrc** (configuration file): reveals bridges and connection settings
- **cached-certs / cached-microdescs:** timestamps = last network connection
- Method: baseline BEFORE, diff AFTER - work on **copies**, hash everything

Bridges: Hiding the Hiding

When networks block Tor, bridges - hidden entry relays - take over:

- Not on public lists, so networks **can't block them**
- Tor Browser suggests them automatically when normal connection fails
- Investigator clue: **"Bridge" lines** in the torrc file on a seized machine
- Bridge used = the suspect knew how to hide even the Tor connection

How Anonymity Breaks: 3 Groups

Tor is strong, but humans are weak. Big agencies break anonymity three ways:

- **OPERATIONAL mistakes:** real email on a forum, same username everywhere - Silk Road fell this way
- **TECHNICAL mistakes:** misconfigured server, code leaking the real IP
- **FINANCIAL trail:** crypto → exchange KYC → real name - money always leaves a trace
- Trainee focus: group 1 - small everyday mistakes solve the most cases

Crypto: The Payment of the Dark Web

Dark markets run on cryptocurrency. Three things you must understand:

- **WALLET:** an address like an account number - but no name on it
- **BLOCKCHAIN:** every transaction public forever - amounts visible, names hidden
- **EXCHANGE:** where crypto becomes rupees - **KYC records live HERE**
- BNSS s.94 notice to the exchange = the real name. Module 10 covers this fully.

Indian Laws for Dark-Web Cases

The same laws apply on the dark web as on the street:

- **NDPS Act:** darknet drug orders - commercial quantity = 10 years to life
- **IT Act ss.43/66:** stolen-data trade and hacking
- **BNSS s.94:** demand records from anyone - your most-used weapon
- **BSA s.63:** electronic records need a certificate to stand in court

Limits for the Investigator

Three limits bind every dark-web investigator. Respect them:

- **LAW:** observe only, never join - buying or selling means YOU commit the crime too
- **SAFETY:** never from your real machine, never without preparation
- **AUTHORITY:** deep investigation belongs to NCB/CBI - your strength is planning, documentation, coordination
- And remember: **using Tor alone is NOT a crime**

Tor Is Not a Crime Tool

Most Tor users are ordinary people protecting themselves:

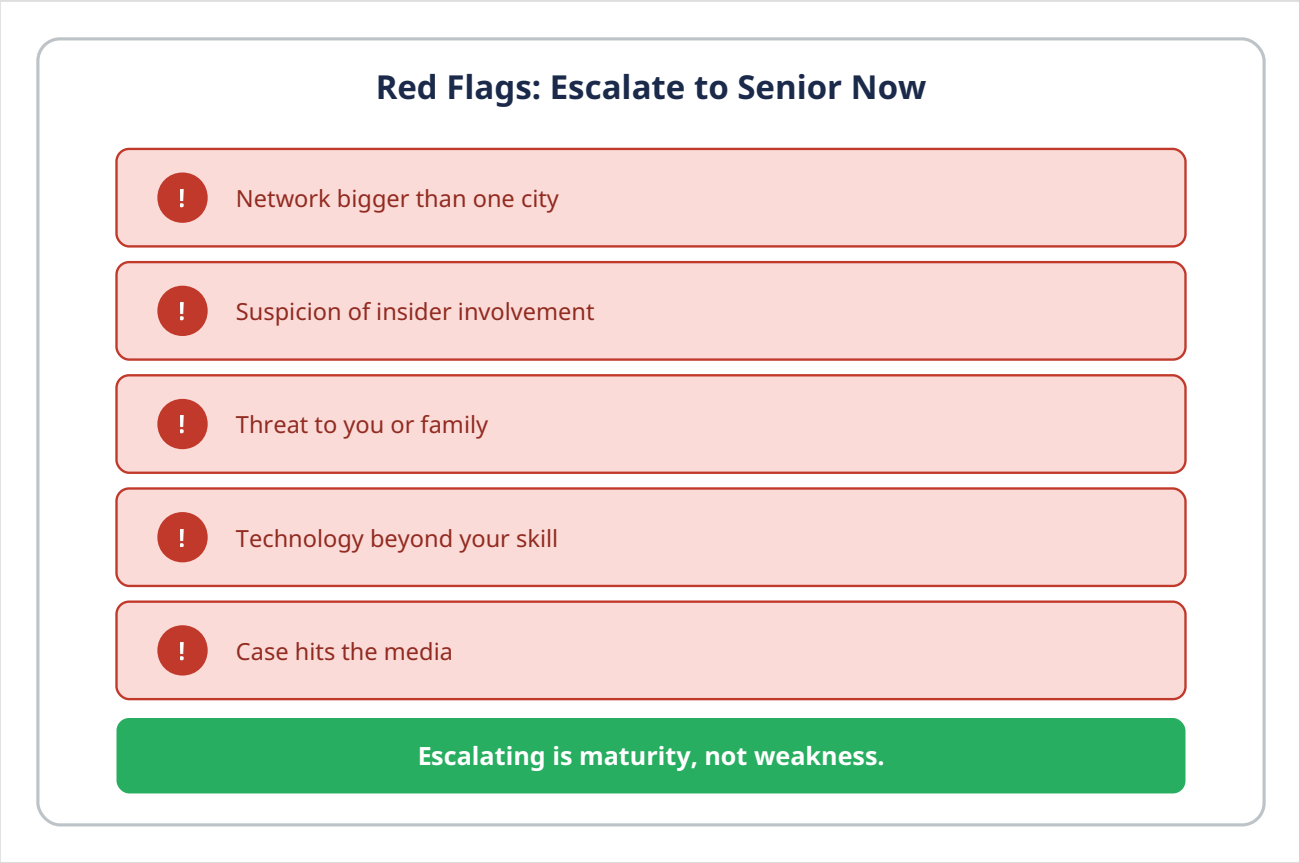
- Journalists on sensitive stories, RTI activists, whistleblowers, censorship-country citizens
- Suspect pattern = Tor **+** crypto transactions **+** suspicious parcels - the whole picture
- **Tor ALONE proves nothing** - never judge on one data point

Open-Source Monitoring: Trainee-Safe

What a trainee CAN do - fully safe, fully legal:

- **Google Alerts** on "darknet drugs India" - automated monitoring, zero risk
- **PIB press releases:** NCB seizure patterns, checked once a week
- **Public court orders:** old cases reveal which evidence actually worked
- Weekly one-page summary + logbook - **observer, never participant**

Red Flags: Escalate NOW



- Network bigger than your city? → **NCB/CBI, immediately**
- Threats to you, internal involvement, evidence tampering → senior NOW
- Escalating is professional maturity, not weakness

Patterns of Dark-Web Crime in India

Three patterns repeat again and again. Learn them:

- **Drug parcels:** from abroad, crypto payment - start with COURIER records
- **Data sale:** stolen Indian user data - first find the breach SOURCE, then the seller
- **Fake documents:** passports, degrees - follow the PAYMENT trail
- Common thread: online traces + crypto + real-world delivery - join all three

Real Case: Silk Road and Ross Ulbricht

The biggest darknet market, 2011-2013. Its fall is a masterclass in OPSEC failure:

- Ross Ulbricht ran Silk Road as **"Dread Pirate Roberts"** - Bitcoin payments
- FBI arrested him in 2013 in a San Francisco library - **laptop open, logged into the admin panel**
- Early mistakes: real email on a forum, identity-linked posts
- Sentence: **life imprisonment** - one small mistake brought down the empire
- Source FBI, via Reuters

Real Case: NCB's Darknet Seizures

NCB's darknet drug cases follow one repeatable pattern:

- Buyer orders on market → **Bitcoin payment** → parcel from abroad
- NCB joined THREE links: Tor traces on device + blockchain transactions + courier records
- **No single link was enough** - all three together made the case
- Lesson: the financial trail is the strongest link
- Source Press Information Bureau

Your Turn: Labs in This Module

Three labs - all safe, all legal. Field work stays with seniors:

- **Lab 1:** install Tor Browser in a VM - signature-verify, study the circuit display
- **Lab 2:** extract Tor artifacts - baseline, diff, torrc, build a timeline
- **Lab 3:** write a one-page intelligence requirement for a fictional parcel case
- Do only what's written - no extra steps, ever

Key Takeaways

If you remember nothing else from Module 9, remember this:

- Surface, deep, dark: **know which web** you're dealing with
- Onion routing: 3 relays, 3 layers - **no single one knows everything**
- Human error beats Tor - operational mistakes solve most cases
- Money trail is king: blockchain → exchange KYC → real name
- Your limits: observe, document, coordinate - **never participate**
- Escalate early. Patience wins big cases.

Review Questions

- 1. Tor Browser is found on a suspect's laptop. Your colleague calls him a darknet criminal. Give four reasons he's wrong - and what you'd check next.
- 2. Explain what the guard relay knows and doesn't know. Why can't it identify the destination?
- 3. Write the five sections of an intelligence requirement for a darknet parcel case.
- 4. A market uses escrow. How does this change the money trail - and where do you look instead?
- 5. You spot a "Bridge" line in a seized torrc. What does it tell you about the suspect's sophistication?

NEXT: MODULE 10

Cryptocurrency Crime Investigation

Blockchain tracing, mixers & exchange KYC